

ROOFTOP SOLAR PLANTS

A Viable Business Opportunity

Solar rooftop installations are a good investment option considering both tangible and intangible benefits



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For organisations planning to shift from conventional energy to solar power use, a rooftop solar photovoltaic (PV) power plant can not only be a money saver but also money spinner with excess power supplied to the utility grid. While the Ministry of New and Renewable Energy (MNRE) is still in the process of laying down specifications for incentives, experts feel that with the right policies and execution, solar rooftop installations can be a hot trend in green technology. It is a profitable business concept, and hence a viable investment option.

Installation and requirements

When solar PV modules are installed on a building's rooftop to generate solar power, it is called a rooftop power plant. Rooftop PV installation can either be done for standalone use or to feed into the grid.

Some of the factors to consider before installing a solar power plant on your building's rooftop include electrical load, current rate, roof size, load capacity and geographic location of the building. The subsidy given by the central and state governments, local utilities, and local community regulations and incentives are also some key determinants in the evaluation.

Rooftop solar arrays are best installed on a large and flat roof where direct sunlight without shadow from the surrounding structures is available. If there is shadow on a part of the terrace during the day, PV solar panels are unable to harvest the sun's energy for that period of time. Let us look at the key considerations while evaluating solar rooftop options.

First, it is important to have a basic understanding of the components of a solar power system and how these generate electricity. PV solar power systems are very simple electric power generating systems comprising the following basic components:

1. A set of PV panels that convert sunlight (photons) into direct-current (DC) electricity
2. A racking system that firmly holds the panels to the roof, exposing these to the sun at an advantageous angle
3. Inverters that convert DC electricity into alternating current (AC) electricity
4. Wiring that connects everything
5. A storage battery (in the case of a grid-fed power plant, a large-sized battery is not necessary to store and use that power after sunset)
6. A variety of means to tilt the panels toward the sun to generate more electricity
7. Energy meters to record the energy that is supplied to the grid
8. Junction boxes
9. Earthing kits

Currently, commercially available silicon-based solar PV panels are made from solar cells encased in a special